B.7.9 PostScript Output

`%!`	standard PostScript file beginning
`MathPictureStart`	initialization of *Mathematica* graphics state
`%%Creator: Mathematica`	standard PostScript convention
`%%AspectRatio:` *r*	the "suggested" ratio of height to width for the final PostScript image
`%%Identifier:` *text*	text from the *Mathematica* input line that generated this output
objects `MathScale`	PostScript computation for scaling of the final picture
`% Start of Graphics`	marker for the beginning of actual graphics
graphics	PostScript for actual *Mathematica* graphics
`% End of Graphics`	marker for the end of actual graphics
`MathPictureEnd`	termination of *Mathematica* graphics

Typical sequence of PostScript generated by *Mathematica*.

The PostScript output generated directly by the *Mathematica* kernel uses a number of special PostScript procedures. Most *Mathematica* front ends automatically interpret these special procedures. However, if you want to send PostScript from *Mathematica* to a generic PostScript output device, you will have to include definition of the procedures, written in PostScript. On most systems, the shell script `psfix` inserts the necessary definitions, and also adds PostScript commands to set up output for $8\frac{1}{2} \times 11$ inch pages, and to specify standard fonts.

The treatment of text is the most complicated aspect of PostScript output from *Mathematica*.

The basic problem is that different devices have different styles and sizes of fonts available. *Mathematica* functions like `FontForm` allow you to give strings which specify fonts for particular pieces of output. Typically these strings are generic font specifications, such as `Plain`, `Italic` or `Bold`. They have to be mapped into the particular fonts that are available on the output device you use. If your output goes through a *Mathematica* front end, the front end will usually take care of the necessary mappings. For output to generic PostScript devices, the mapping must be done by an external program such as `psfix`. You may well have to modify `psfix`

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to deal with different sets of fonts on different output device has.

A second complication concerns the size of text. Although some PostScript fonts can be in principle be scaled arbitrarily, actual finite-resolution output devices typically use a discrete set of different-sized fonts. This means that the size of a particular piece of text can be determined only when the PostScript for it is actually being rendered. As a result, the precise position the text, and, in fact, the overall scaling of the whole picture, must be determined during PostScript rendering, by executing the special PostScript procedure `MathScale`.

The graphics output produced by *Mathematica* can be rendered on any device that supports a full version of standard PostScript. The definitions for the special PostScript procedures used in *Mathematica* output are sufficiently complicated that few subsets of PostScript will be adequate.

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